

- 1** A solid iron sphere of density 8000 kg m^{-3} and volume $4.50 \times 10^{-4} \text{ m}^3$ is completely submerged in a liquid of density 800 kg m^{-3} . The iron sphere is resting on a spring, as shown in Fig. 1.1. The spring is compressed by 10.2 cm.

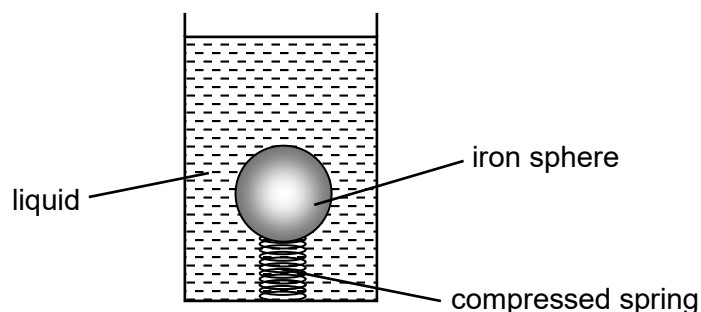


Fig. 1.1

- (a)** Show that the upthrust on the iron sphere is 3.53 N.

[1]

Solution

$$\begin{aligned} \text{Upthrust} &= \rho V g \\ &= 800(4.50 \times 10^{-4})(9.81) \\ &= 3.5316 \text{ N} \\ &= 3.53 \text{ N (Shown)} \end{aligned}$$

M1

A0

- (b)** Hence, calculate the force constant of the spring.

force constant = N m^{-1} [2]

Solution

At equilibrium, considering forces on the iron sphere

$$kx + U = mg$$

$$\begin{aligned} k &= \frac{mg - U}{x} = \frac{8000(4.50 \times 10^{-4})(9.81) - 3.5316}{0.102} \\ &= 312 \text{ N m}^{-1} \end{aligned}$$

M1

A1

- (c) A string of breaking strength 32.0 N is used to lift the iron sphere vertically upwards, as shown in Fig. 1.2. The iron sphere is then lifted partially out of the liquid as shown in Fig. 1.3.

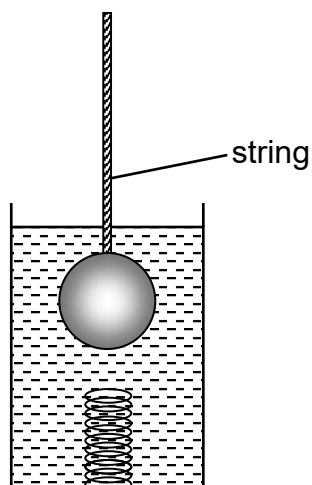


Fig. 1.2

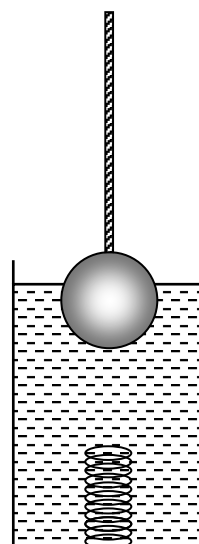


Fig. 1.3

- (i) Explain why the string breaks as the sphere emerges from the liquid.

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[2]

Solution

When the sphere is lifted out of the liquid, the volume of liquid displaced is reduced. This causes the upthrust acting on sphere to decrease.

B1

To maintain equilibrium, the tension will increase and the string breaks when the tension exceeds the maximum allowable value.

B1

- (ii) Calculate the volume of the fluid displaced at the instant when the string breaks.

volume = m³ [3]

Solution

At breaking point,
 $T + U = mg$

$$32.0 + 800(V)(9.81) = 8000(4.50 \times 10^{-4})(9.81)$$

$$V = 4.23 \times 10^{-4} \text{ m}^3$$

C1
M1
A1

[Total: 8]

[Turn over

